

1. First Fit

Fragmentation tendency:

First Fit may cause external fragmentation because repeated allocations can leave scattered free spaces of different sizes. However, it generally provides a reasonable balance between memory utilization and allocation speed.

Usefulness in long-running, real-world systems:

Practically, First Fit is often preferred in long-running, real-world systems because it is fast, simple, and requires less searching overhead. However, its effectiveness depends on the workload and fragmentation pattern.

2. Best Fit

Fragmentation tendency:

Best Fit minimizes immediate memory wastage by selecting the smallest suitable partition. However, repeated allocations may create many small unusable free spaces, leading to external fragmentation.

Usefulness in long-running, real-world systems:

Theoretically, Best Fit may provide better memory utilization by minimizing immediate wastage. However, practically, its higher searching overhead and the possibility of creating many small unusable fragments may reduce its effectiveness in long-running systems. Its usefulness ultimately depends on the workload and fragmentation pattern.

3. Worst Fit

Fragmentation tendency:

Worst Fit allocates a process to the largest available partition, leaving relatively large remaining holes. However, repeated allocations may consume large partitions and result in poor memory utilization and higher memory wastage.

Usefulness in long-running, real-world systems:

Worst Fit is generally less preferred in long-running, real-world systems because it may consume large partitions unnecessarily, reducing memory utilization and making it harder to accommodate future processes. However, its effectiveness also depends on the workload.

★ Final Comparison — Easy to Remember

Algorithm	Fragmentation Tendency	Long-Running / Real-World Usefulness
First Fit	Scattered free spaces → may cause external fragmentation	★ Often practically preferred because it is fast and simple
Best Fit	May create many small unusable fragments	Theoretically good for memory utilization, but higher search overhead
Worst Fit	May cause poor utilization and memory wastage	Generally less preferred

□ One-line memory trick

First Fit → Practically Fast

Best Fit → Theoretically Space-Efficient

Worst Fit → Generally Less Preferred

আর সবচেয়ে গুরুত্বপূর্ণ concluding line:

There is no universally best placement algorithm; the most suitable algorithm depends on the workload, allocation pattern, fragmentation, and performance requirements.

First Fit, Best Fit & Worst Fit — Theoretical vs Practical

1. Best Fit — Theoretically Better

Theoretically, Best Fit is better because it selects the **smallest suitable partition** for each process. Therefore, it reduces immediate memory wastage, improves memory utilization, and keeps larger partitions available for larger processes.

However, Best Fit requires scanning more partitions, so it has **higher searching overhead**.

Exam Line:

Theoretically, Best Fit is better because it minimizes immediate memory wastage and provides better memory utilization.

2. First Fit — Practically Better

Practically, First Fit is generally better because it is fast, simple, and requires less searching. It allocates a process to the first suitable partition instead of searching the entire memory.

Although its memory utilization may not be as good as Best Fit, its lower searching overhead and faster allocation make it more practical for real-world and long-running systems.

Exam Line:

Practically, First Fit is better because it is faster, simpler, and has lower searching overhead, making it more suitable for real-world systems.

3. Worst Fit — Generally Less Preferred

Theoretically and practically, Worst Fit is generally less preferred because it allocates a process to the largest available partition. Although it leaves relatively large remaining holes, it may consume large partitions unnecessarily and result in poor memory utilization.

Exam Line:

Worst Fit is generally less preferred because it may cause poor memory utilization and higher memory wastage despite leaving relatively large free holes.

★ সবচেয়ে গুরুত্বপূর্ণ Comparison

বিষয়	Best Fit	First Fit	Worst Fit
Theoretically Better	✓ Yes	—	✗
Practically Better	—	✓ Yes	✗
Main advantage	Better memory utilization	Fast & simple	Leaves larger remaining holes
Searching	বেশি	কম	বেশি
Fragmentation	Small fragments তৈরি হতে পারে	Moderate	Large blocks দ্রুত consume হতে পারে

বিষয়	Best Fit	First Fit	Worst Fit
Real-world usefulness	Moderate	☆ High	Low

🔥 Exam-এ সরাসরি এইভাবে লিখতে পারো:

Theoretically, Best Fit is considered better because it selects the smallest suitable partition, which reduces immediate memory wastage and improves memory utilization. However, practically, First Fit is generally preferred because it is faster, simpler, and requires less searching overhead. Worst Fit is generally less preferred because it may lead to poor memory utilization and higher memory wastage.

1. First Fit

Advantages

- Allocation is very fast.
- It allocates a process to the first suitable partition.
- It requires less memory scanning.
- It is simple and easy to implement.

Disadvantages

- Memory utilization is not always efficient.
- Large partitions may become occupied by small processes.
- Suitable partitions may not be available for larger processes later.
- It may cause internal fragmentation.

Exam Line

First Fit is fast and easy to implement, but it may lead to poor memory utilization and internal fragmentation.

2. Best Fit

Advantages

- It selects the smallest suitable partition for a process.
- Memory wastage is comparatively lower.

- It provides better memory utilization.
- It increases the possibility of allocating more processes.

Disadvantages

- It requires scanning all partitions before allocation.
- Allocation takes more time.
- It is comparatively more complex to implement.
- It has higher searching overhead.

Exam Line

Best Fit provides better memory utilization and less memory wastage, but it requires scanning all partitions and takes more time.

3. Worst Fit

Advantages

- It allocates a process to the largest available partition.
- By using the largest partition, medium-sized partitions may remain available for future processes.
- Its allocation logic is simple.

Disadvantages

- Large partitions become occupied quickly.
- Memory utilization may be poor.
- It requires scanning all partitions.
- It may fail to allocate a larger number of processes.
- It may cause higher internal fragmentation.

Exam Line

Worst Fit allocates the largest partition first, but it often results in poor memory utilization and higher memory wastage.

Best Fit is the Best Placement Algorithm

Best Fit is considered the best placement algorithm because it allocates each process to the smallest suitable memory partition. As a result, large partitions remain available for larger processes, which improves memory utilization and reduces memory wastage. Therefore, more processes can be allocated successfully than with First Fit or Worst Fit.

Although Best Fit requires scanning all memory partitions and takes more time, it provides the most efficient use of memory. Hence, Best Fit is generally considered the best placement algorithm.

Key Reasons

- **Selects the smallest suitable partition for each process.**
- **Reduces memory wastage.**
- **Improves memory utilization.**
- **Keeps larger partitions available for larger processes.**
- **Increases the chance of allocating more processes.**

Very Short Exam Answer (2 Marks)

Best Fit is the best placement algorithm because it always selects the smallest suitable partition for a process. This reduces memory wastage, improves memory utilization, keeps larger partitions available for larger processes, and increases the possibility of allocating more processes successfully.